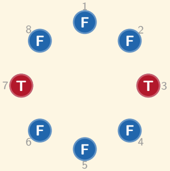
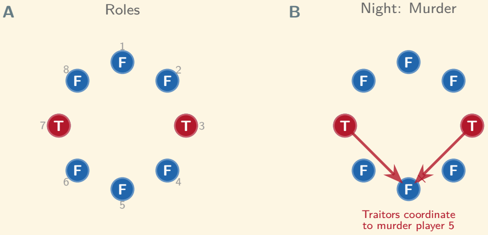
The Vote-Left Equilibrium: A Deterministic Coordination Strategy for the Faithful in The Traitors

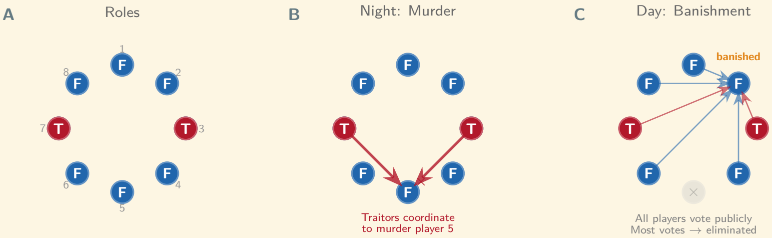
Vince Knight, Cardiff

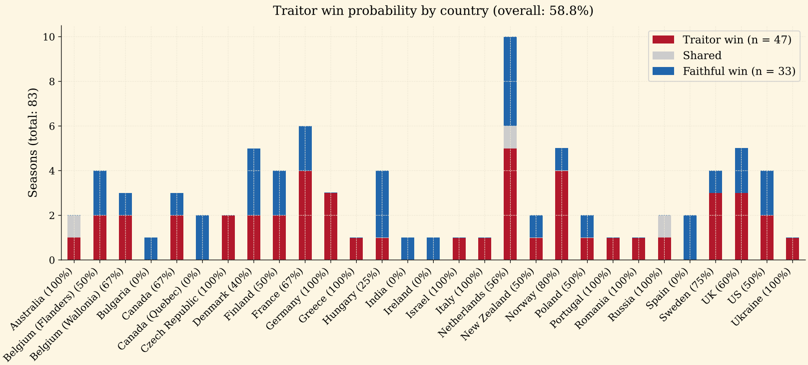
A

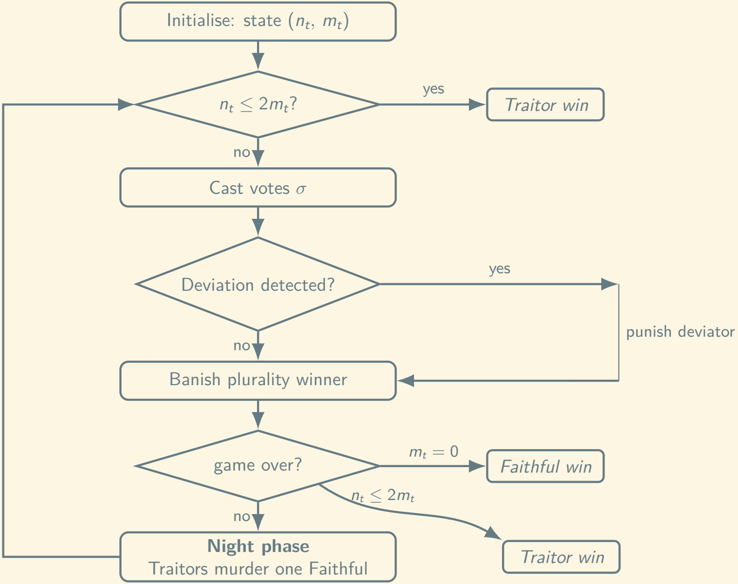
Roles



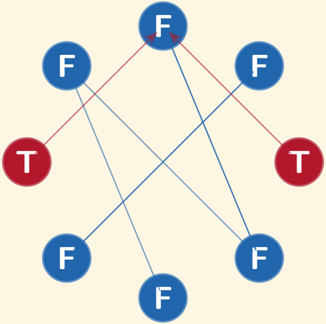

A
Roles
1
2
3
4
5
6
7
8
F
T
B
Night: Murder
Traitors coordinate
to murder player 5


A
Roles
1
2
3
4
5
6
7
8
F
T
B
Night: Murder
Traitors coordinate
to murder player 5
C
Day: Banishment
banished
All players vote publicly
Most votes → eliminated


Traitor win probability by country (overall: 58.8%)
Traitor win (n = 47)
Shared
Faithful win (n = 33)
Seasons (total: 83)
Australia (100%)
Belgium (Flanders) (50%)
Belgium (Wallonia) (67%)
Bulgaria (0%)
Canada (67%)
Canada (Quebec) (0%)
Czech Republic (100%)
Denmark (40%)
Finland (50%)
France (67%)
Germany (100%)
Greece (100%)
Hungary (25%)
India (0%)
Ireland (0%)
Israel (100%)
Italy (100%)
Netherlands (56%)
New Zealand (50%)
Norway (80%)
Poland (50%)
Portugal (100%)
Romania (100%)
Russia (100%)
Spain (0%)
Sweden (75%)
UK (60%)
US (50%)
Ukraine (100%)


Initialise: state (n_t, m_t)
$n_t \leq 2m_t$?
yes
Traitor win
no
Cast votes σ
Deviation detected?
yes
punish deviator
no
Banish plurality winner
game over?
$m_t = 0$
Faithful win
$n_t \leq 2m_t$
Traitor win
no
Night phase
Traitors murder one Faithful

Random Voting

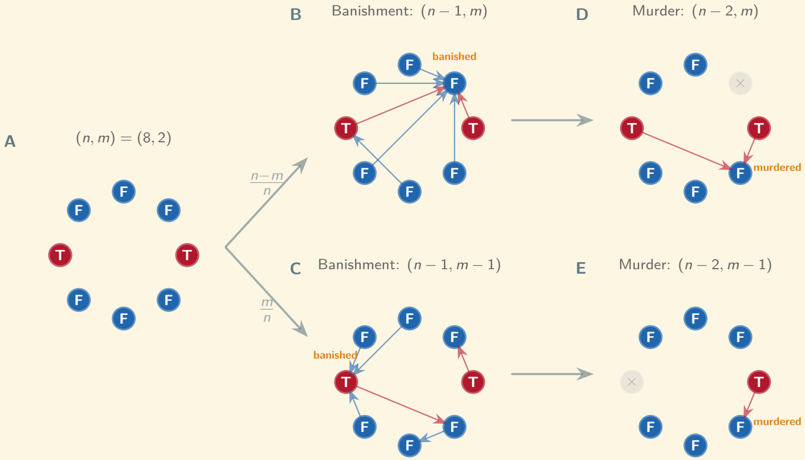


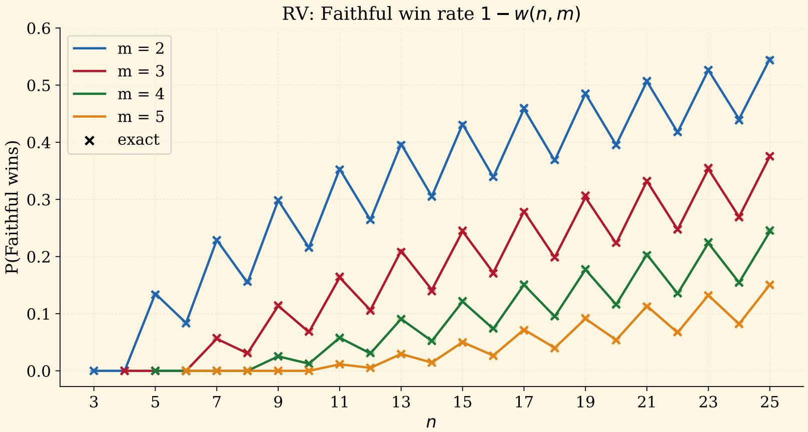
Proposition (Migdał[1] recurrence, adapted)

Let $w(n, m)$ denote the Traitor win probability under mutual random play from state (n, m). Then

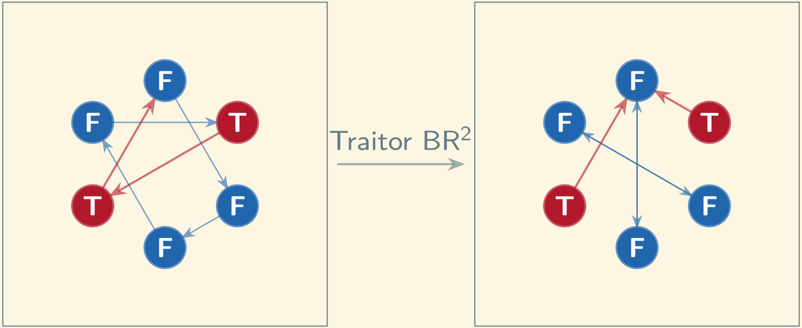
$$w(n,m) = \begin{cases} 0 & \text{if } m = 0, \\ 1 & \text{if } n \le 2m, \\ \dfrac{n-m}{n}\, w(n-2, m) \;+\; \dfrac{m}{n}\, w(n-2, m-1) & \text{if } n > 2m. \end{cases}$$

[1] Piotr Migdał. *A mathematical model of the Mafia game*. arXiv:1009.1031. 2010.


A
$(n, m) = (8, 2)$
$\frac{n-m}{n}$
$\frac{m}{n}$
B
Banishment: $(n-1, m)$
banished
C
Banishment: $(n-1, m-1)$
banished
D
Murder: $(n-2, m)$
murdered
E
Murder: $(n-2, m-1)$
murdered
F
T



RV: Faithful win rate $1 - w(n, m)$
P(Faithful wins)
n
m = 2
m = 3
m = 4
m = 5
exact
0.0
0.1
0.2
0.3
0.4
0.5
0.6
3
5
7
9
11
13
15
17
19
21
23
25



RV RV+C

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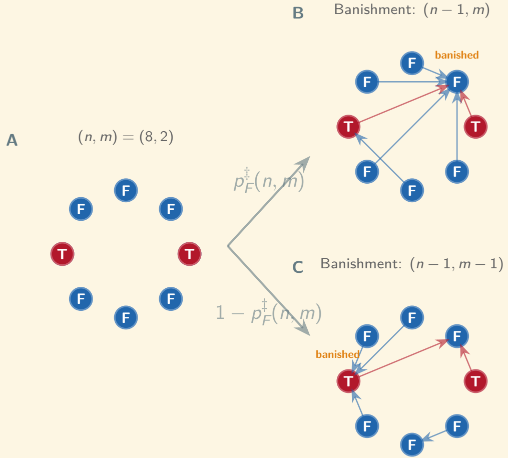
[2] Sizhe Wang. "Optimal strategy in the Werewolf game: A theoretical study". In: *Games: Research and Practice* (2024). arXiv:2408.17177.

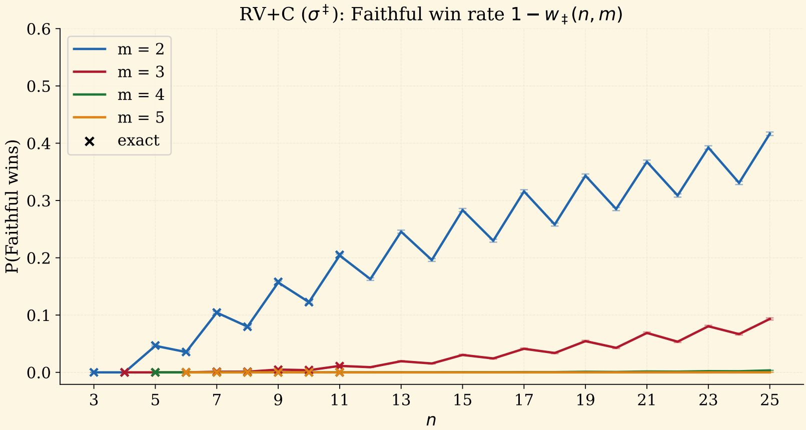

A
$(n, m) = (8, 2)$
B
Banishment: $(n-1, m)$
banished
C
Banishment: $(n-1, m-1)$
banished
$p_F^{\ddagger}(n, m)$
$1 - p_F^{\ddagger}(n, m)$
F
T

Lemma (Collusion banishment probability)

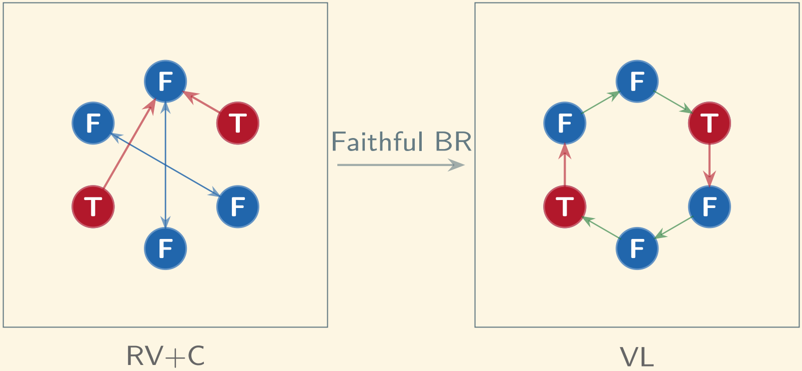
The probability that the eliminated player is Faithful under $\sigma^{\ddagger}$ is

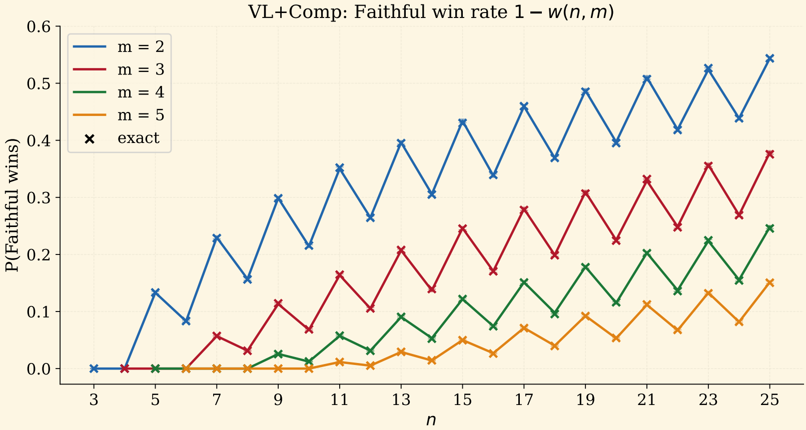
$$p_F^{\ddagger}(n, m) = \frac{1}{|\Omega|} \sum_{v \in \Omega} \frac{|W(v) \cap F|}{|W(v)|},$$

where $W(v) = \{j : C_j(v) = M(v)\}$.



RV+C ($\sigma^{\ddagger}$): Faithful win rate $1 - w_{\ddagger}(n, m)$
P(Faithful wins)
n
m = 2
m = 3
m = 4
m = 5
exact
0.0
0.1
0.2
0.3
0.4
0.5
0.6
3
5
7
9
11
13
15
17
19
21
23
25


F
F
T
T
F
F
Faithful BR
F
F
T
T
F
F
RV+C
VL

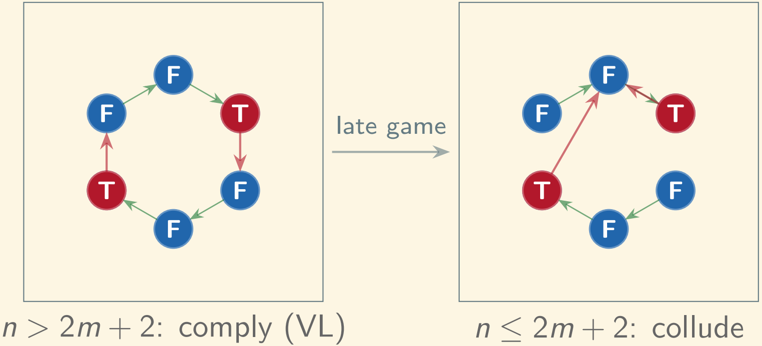


VL+Comp: Faithful win rate $1 - w(n, m)$
P(Faithful wins)
n
m = 2
m = 3
m = 4
m = 5
exact
0.0
0.1
0.2
0.3
0.4
0.5
0.6
3
5
7
9
11
13
15
17
19
21
23
25

Theorem (Bayesian Nash Equilibrium (BNE))

Vote-Left with Punishment constitutes a BNE of $\mathrm{TG}(n, m)$ *for every state with* $n > 2m + 2$.

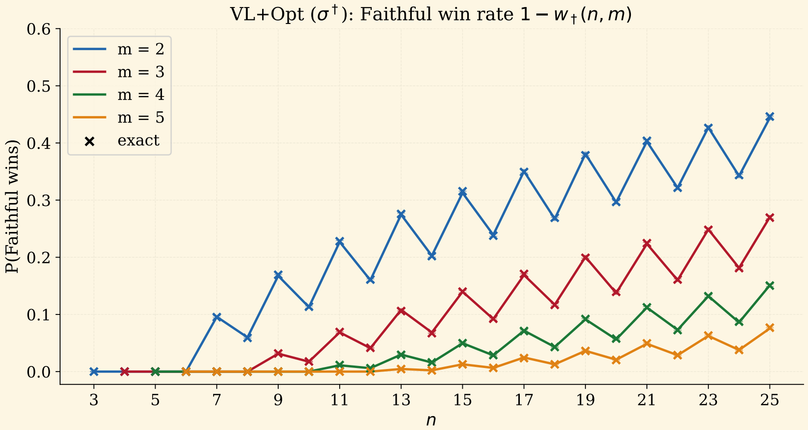
$\sigma^\dagger$ (VL+Opt): the Traitors' best response



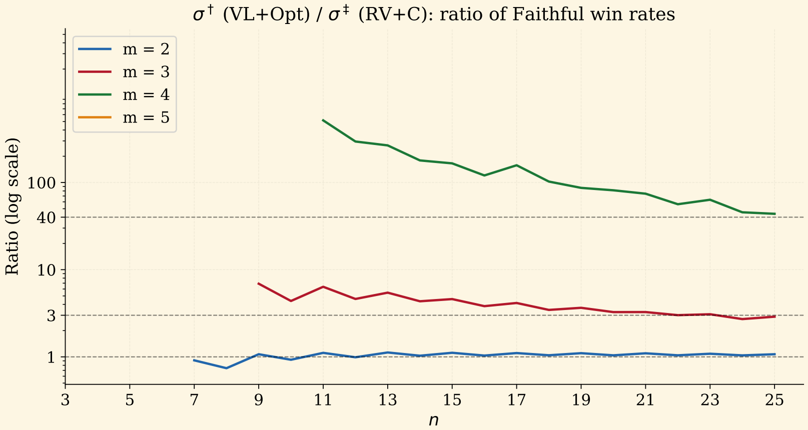
Proposition (Traitor win probability under $\sigma^\dagger$)

Let $w_\dagger(n, m)$ denote the Traitor win probability under Vote-Left with the optimal Traitor strategy $\sigma^\dagger$, starting from state (n, m). Then

$$w_\dagger(n, m) = \begin{cases} 0 & m = 0, \\ 1 & n \le 2m + 2, \\ \dfrac{n-m}{n}\, w_\dagger(n-2,\ m) + \dfrac{m}{n}\, w_\dagger(n-2,\ m-1) & n > 2m + 2. \end{cases}$$


VL+Opt ($\sigma^\dagger$): Faithful win rate $1 - w_\dagger(n, m)$
P(Faithful wins)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
3
5
7
9
11
13
15
17
19
21
23
25
n
m = 2
m = 3
m = 4
m = 5
exact

(n, m)	w	RV	$w_{\ddagger}$	RV+C	$w_{\dagger}$	VL+Opt
$(7, 2)$	0.771	0.771	0.896	0.896	0.905	0.904
$(8, 2)$	0.844	0.846	0.920	0.920	0.941	0.940
$(9, 3)$	0.886	0.886	0.995	0.995	0.968	0.969
$(10, 3)$	0.931	0.932	0.996	0.996	0.982	0.983
$(11, 3)$	0.835	0.837	0.989	0.989	0.931	0.931
$(11, 5)$	0.988	0.989	1.000	1.000	1.000	1.000
$(15, 2)$	0.570	0.568	–	0.717	0.685	0.687
$(20, 3)$	0.776	0.777	–	0.957	0.860	0.863
$(22, 3)$	0.752	0.753	–	0.946	0.839	0.840
$(24, 3)$	0.731	0.731	–	0.933	0.819	0.819
$(25, 3)$	0.625	0.625	–	0.907	0.731	0.730
$(22, 4)$	0.864	0.865	–	0.999	0.927	0.927
$(25, 4)$	0.754	0.757	–	0.997	0.850	0.849



$\sigma^\dagger$ (VL+Opt) / $\sigma^\ddagger$ (RV+C): ratio of Faithful win rates
m = 2
m = 3
m = 4
m = 5
Ratio (log scale)
100
40
10
3
1
3
5
7
9
11
13
15
17
19
21
23
25
n

Vote-Left raises the Faithful's winning probability

by a factor of approximately **three** at $m = 3$

by a factor of **forty or more** at $m = 4$

relative to random voting under Traitor collusion

The Vote-Left Equilibrium

Preprint: `arxiv.org/abs/2605.10233`

Install: `pip install backstab`
Docs: `vknight.org/backstab`
Dev: `github.com/drvinceknight/backstab`

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